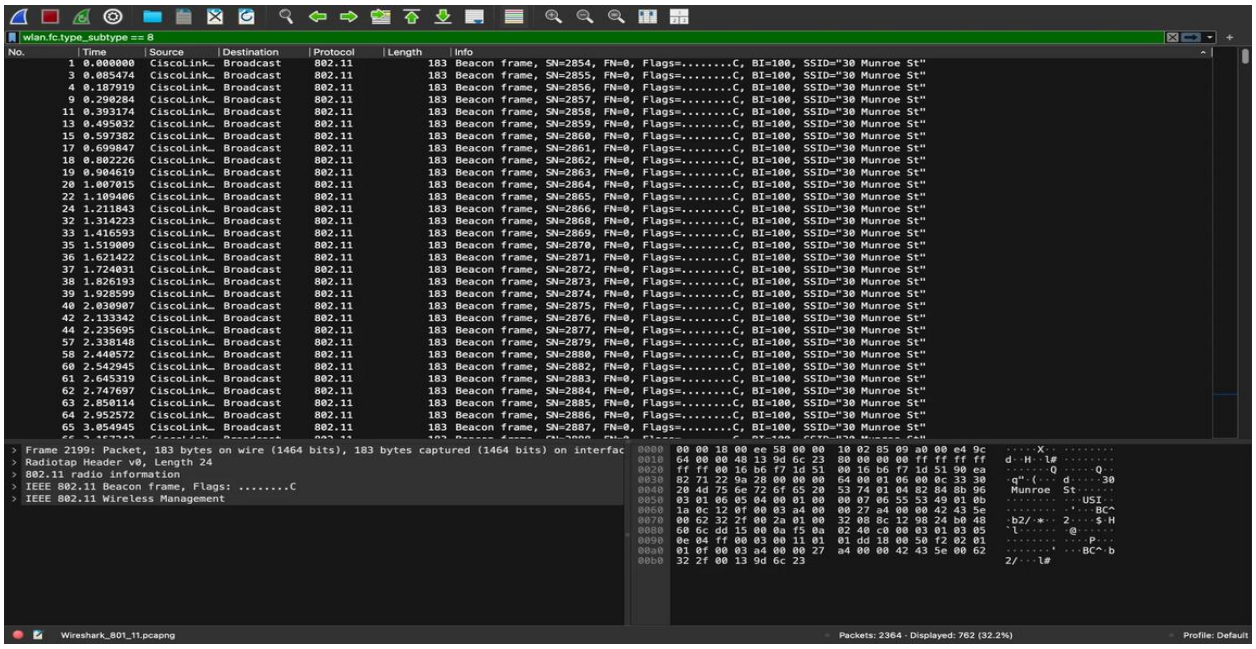
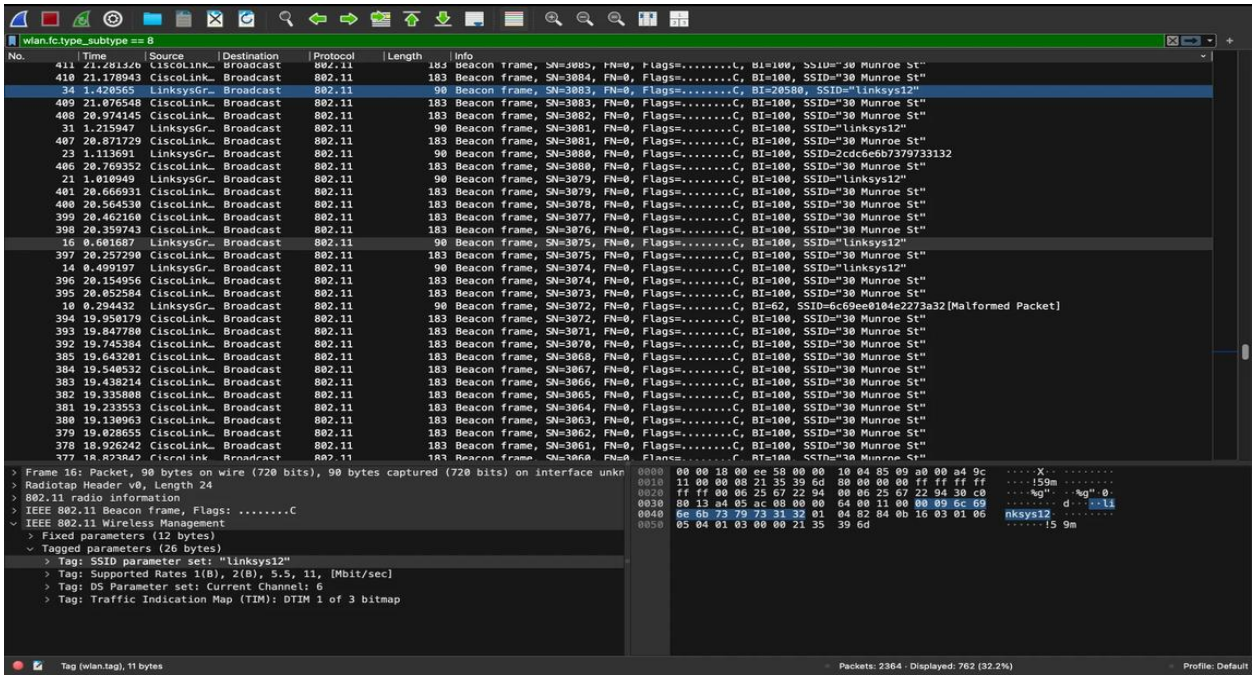
CYSE 425 - Lab 3

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Professor Gebril

Purpose of Lab: The purpose of this lab was to analyze IEEE 802.11 wireless network traffic using Wireshark. By examining captured packets, I investigated beacon frames, MAC and IP addresses, TCP communication, and the processes involved in disconnecting, authenticating, and associating with a wireless access point.

Question 1:

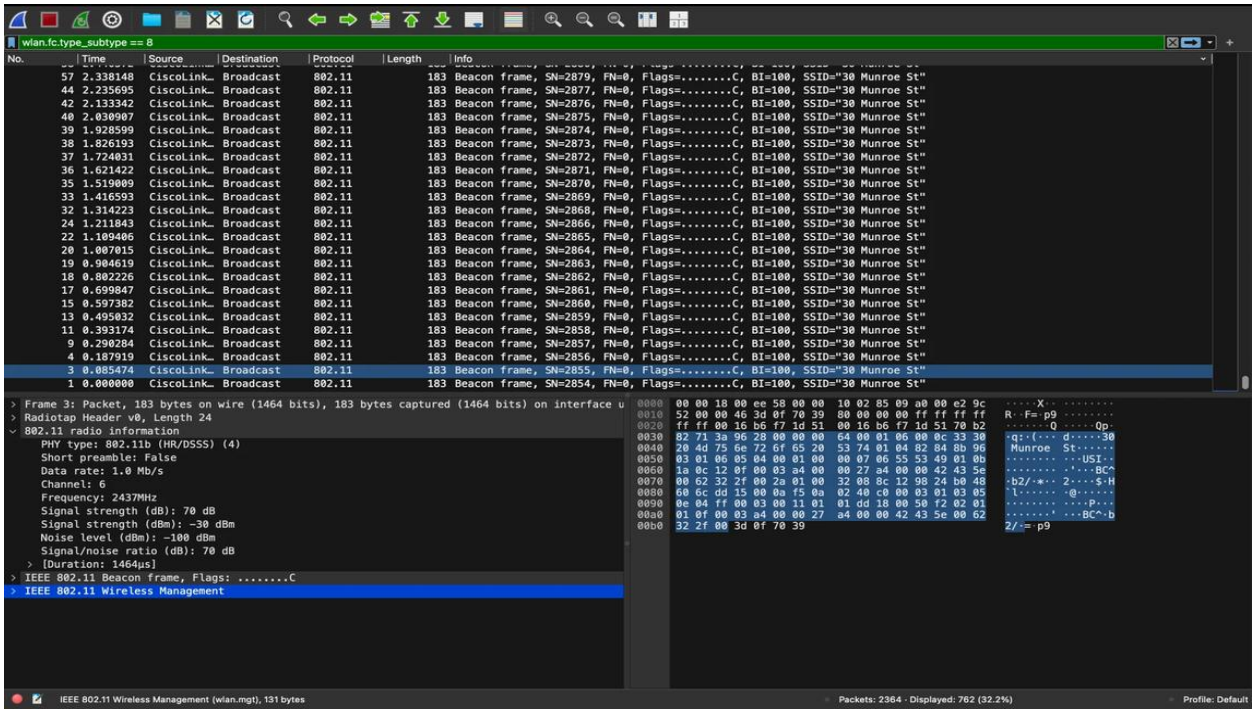


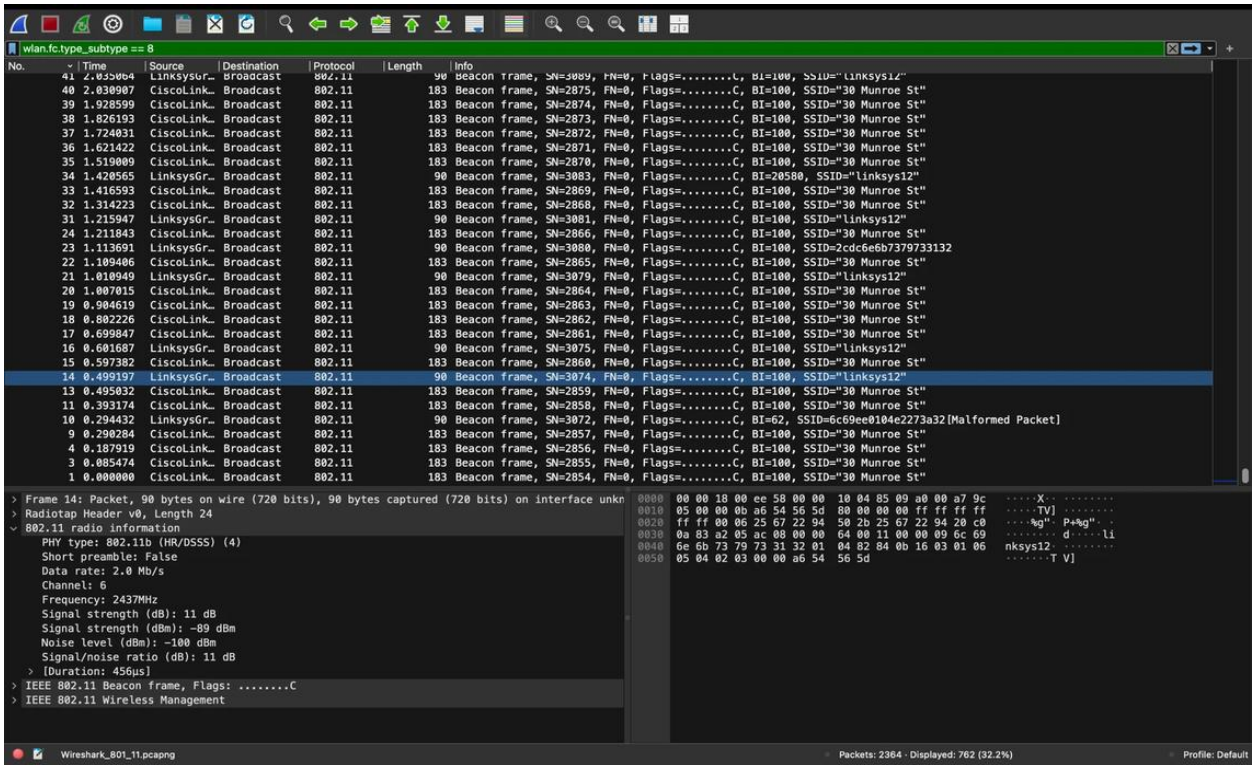


Answer: The two SSIDs issuing the most beacon frames are 30 Munroe St and linksys12

Description: I applied the wlan.fc.type_subtype == 8 display filter to isolate beacon frames. By examining the SSIDs displayed in the Info column, I identified the two wireless networks appearing most frequently in the capture.

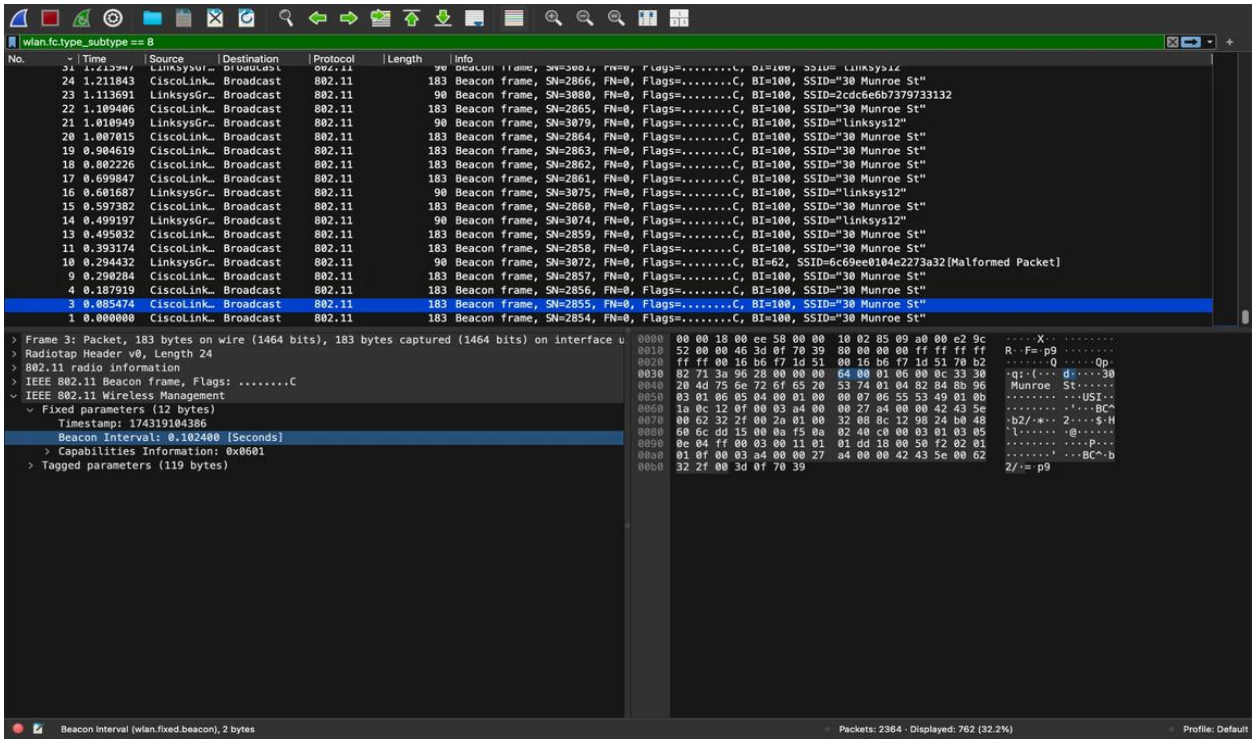
Question 2:





Answer: Both access points, 30 Munroe St and linksys12, operate on Channel 6 (2437 MHz).
Description: I examined the radio information in the beacon frames transmitted by both access points. The Channel field indicated Channel 6, corresponding to a frequency of 2437 MHz.

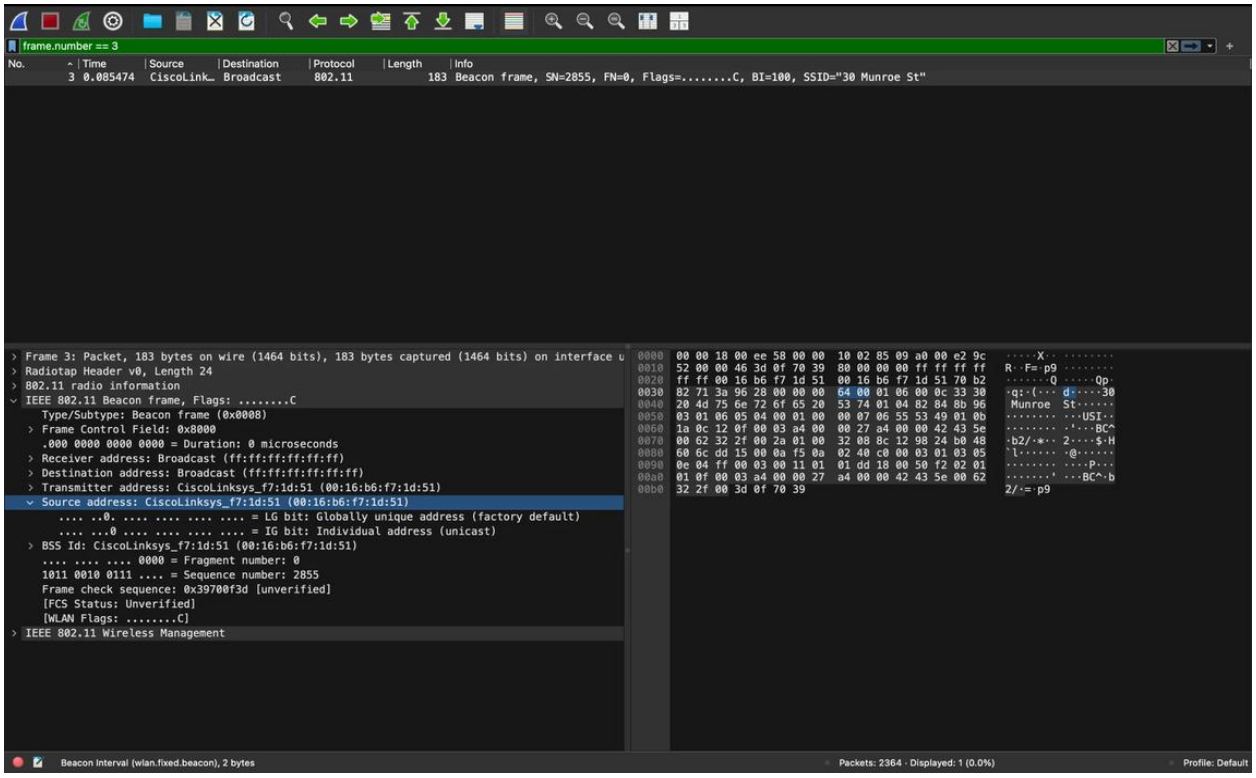
Question 3:



Answer: The beacon transmission interval is 100 Time Units (TUs) which is equivalent to 102.4 milliseconds.

Description: I examined the beacon frame at timestamp 0.085474 seconds and inspected the Beacon Interval field. The interval of 100 TUs indicates that the access point is configured to transmit a beacon approximately every 102.4 milliseconds.

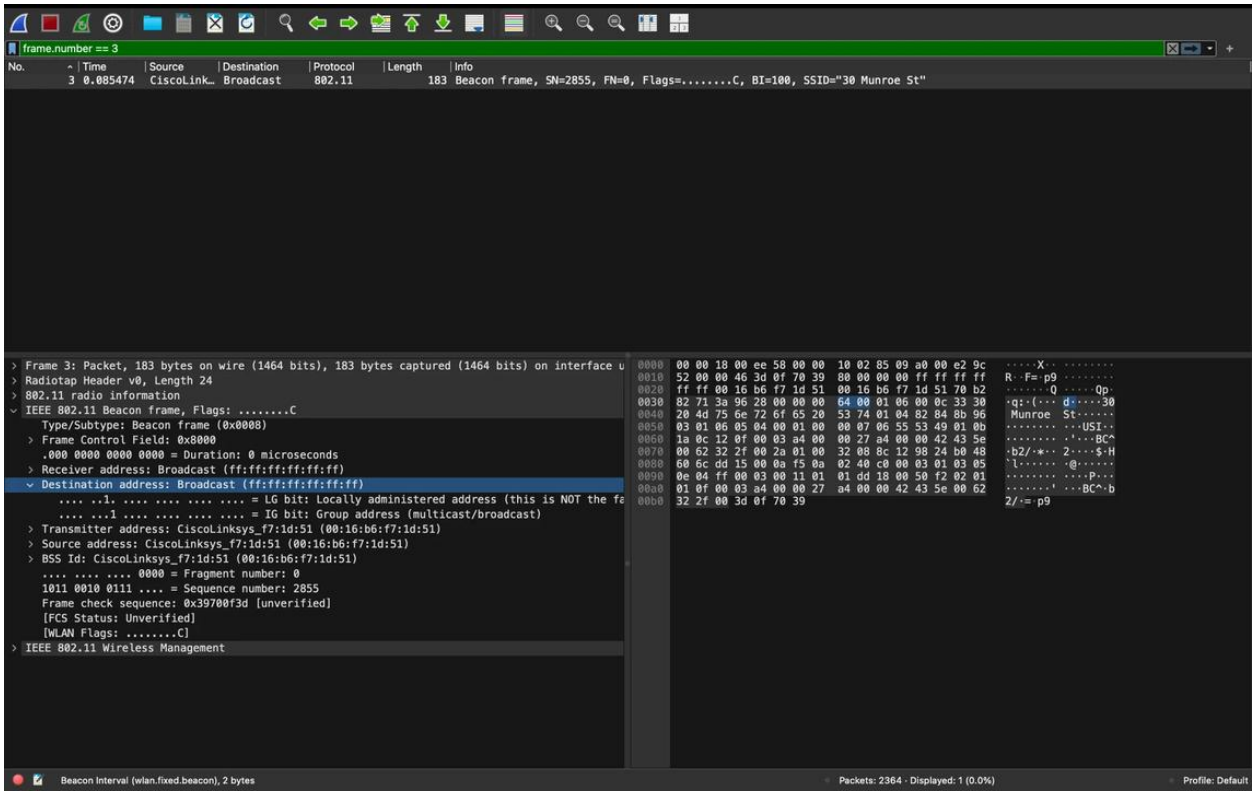
Question 4:



Answer: The source MAC address is 00:16:b6:f7:1d:51.

Description: I examined the beacon frame at timestamp 0.085474 seconds and located the Source Address field within the IEEE 802.11 frame header. This address identifies the 30 Munroe St access point transmitting the beacon.

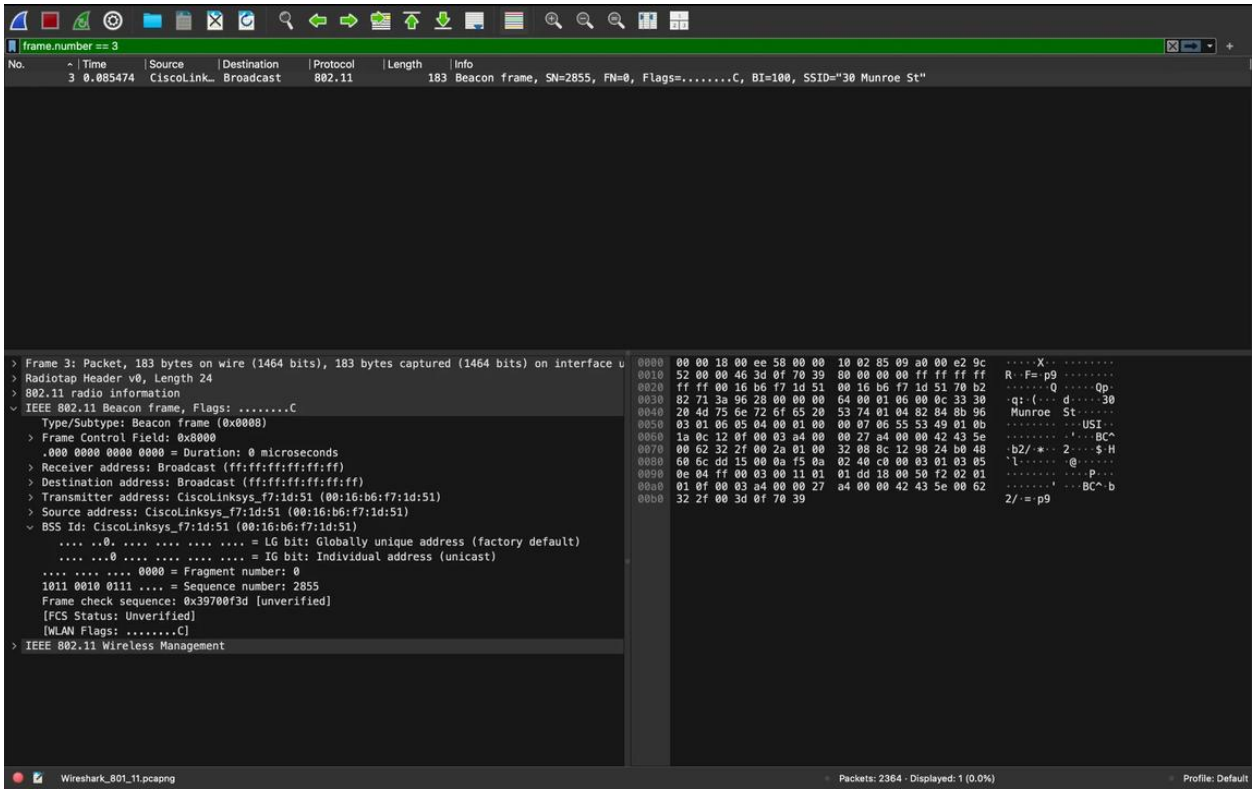
Question 5:



Answer: The destination MAC address is ff:ff:ff:ff:ff:ff.

Description: I examined the Destination Address field in the beacon frame from the 30 Munroe St access point. This broadcast MAC address indicates that the beacon is intended for all nearby wireless devices rather than one specific device.

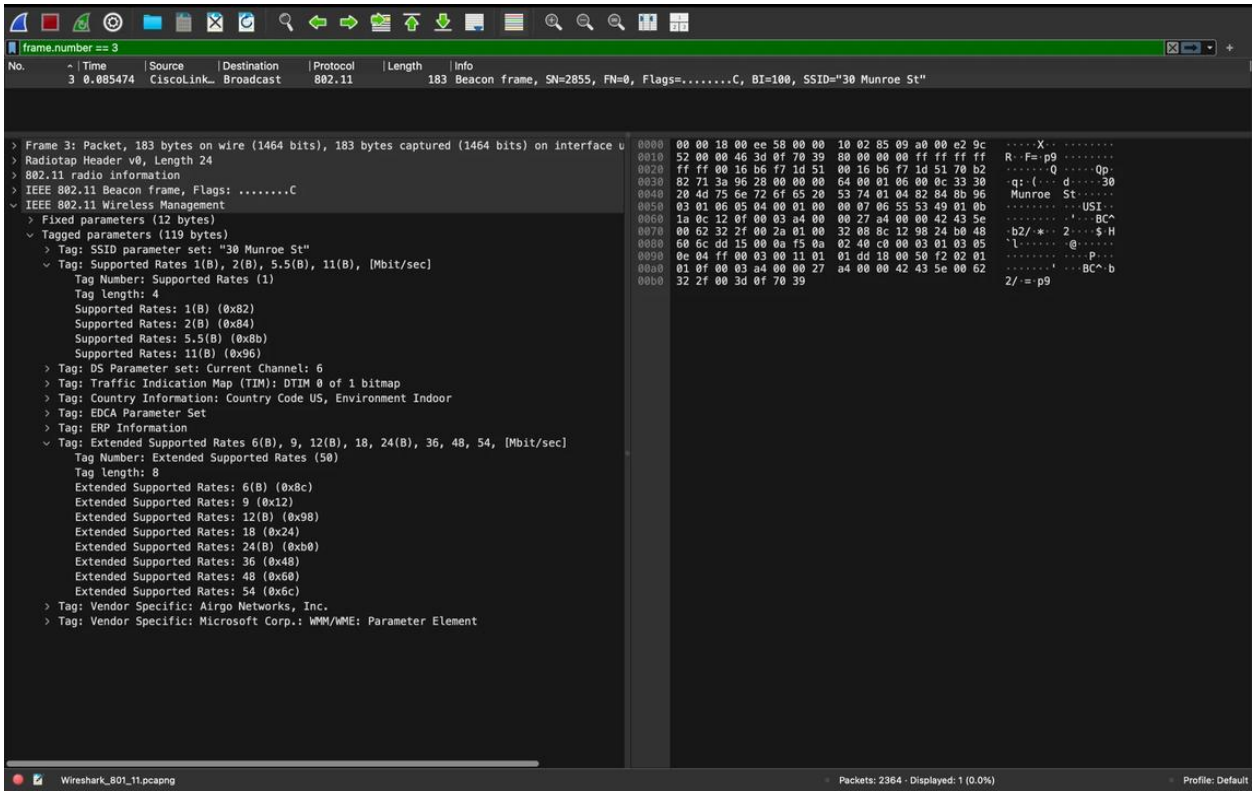
Question 6:



Answer: The BSSID MAC address is 00:16:b6:f7:1d:51.

Description: I examined the BSS ID field within the IEEE 802.11 beacon frame header. The BSSID identifies the wireless basic service set and matches the transmitting access point's MAC address in this frame.

Question 7:

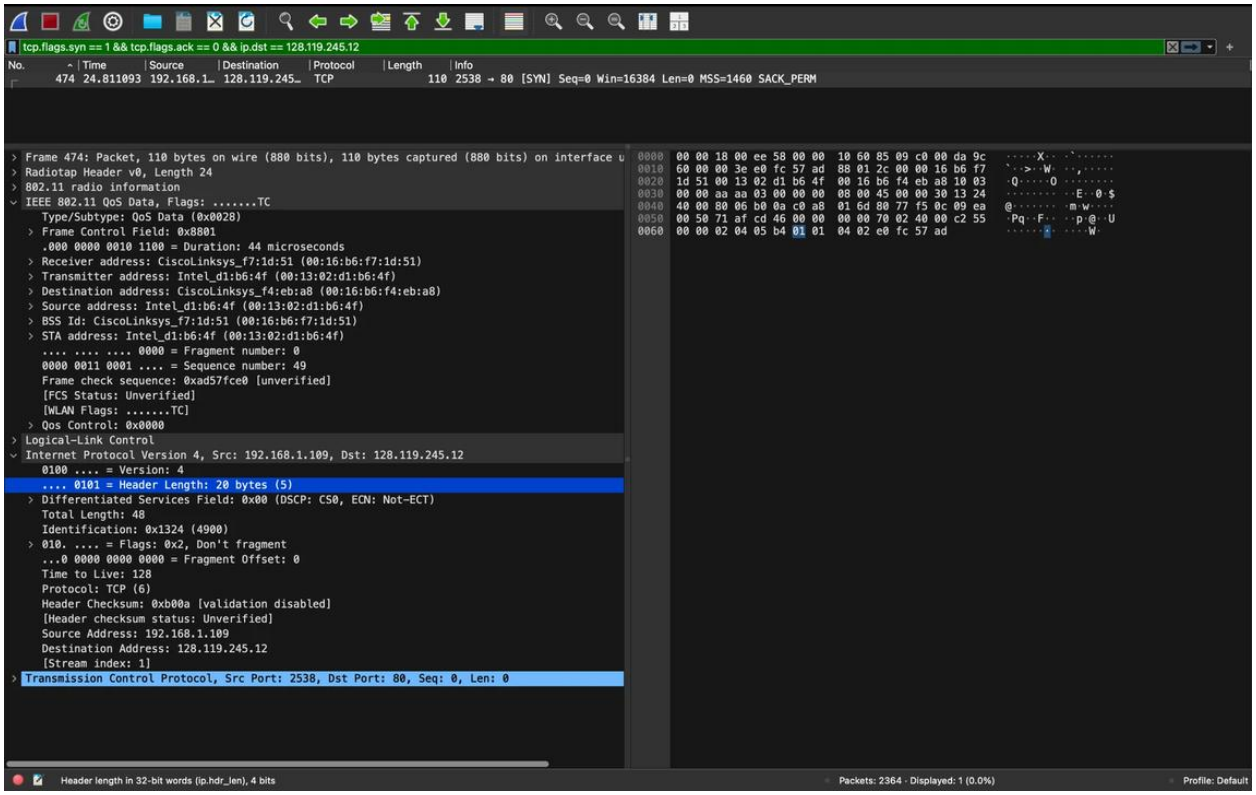


Answer: The access point advertises that:

- Supported Rates: 1, 2, 5.5, and 11 Mbps
- Extended Supported Rates: 6, 9, 12, 18, 24, 36, 48, and 54 Mbps

Description: I examined the beacon frame from the 30 Munroe St access point and expanded the Supported Rates and Extended Supported Rates fields under Tagged parameters. These fields advertise the wireless transmission speeds supported by the access point.

Question 8:

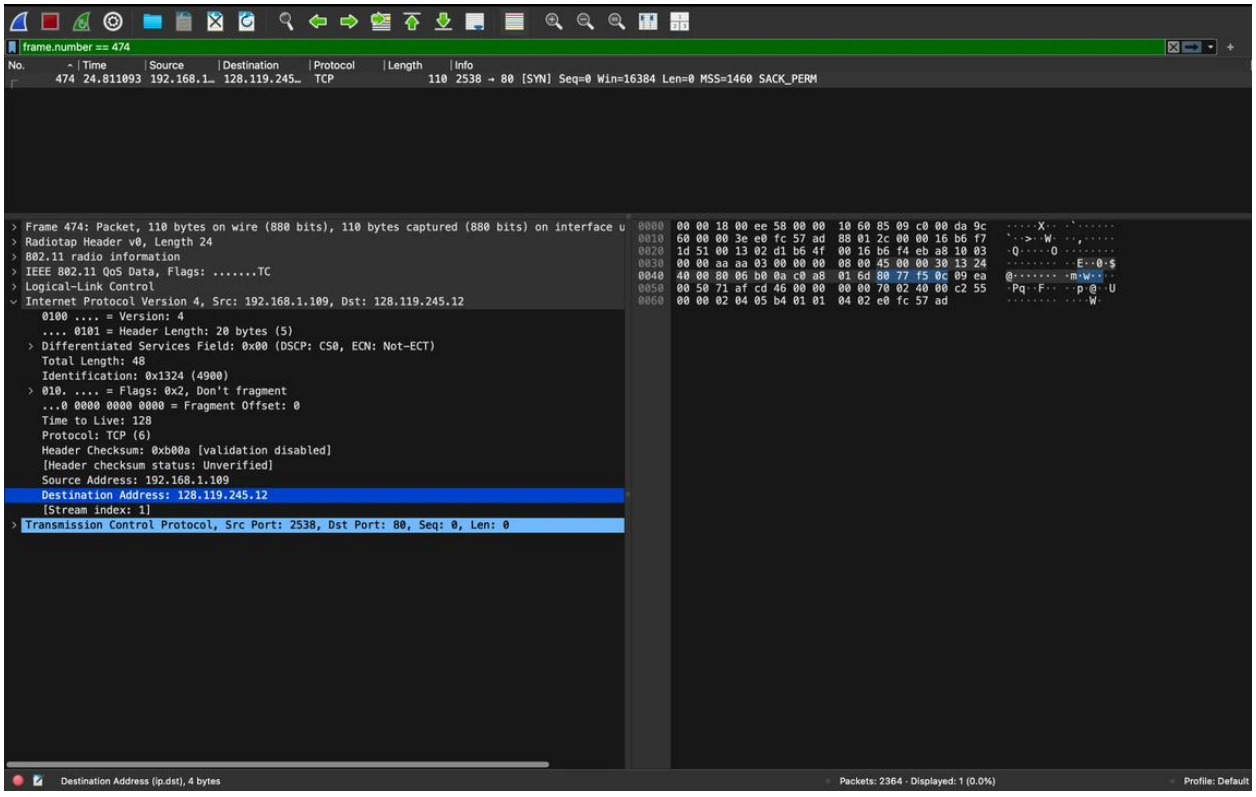


Answer:

- Wireless Host MAC Address: 00:13:02:d1:b6:4f
- Access Point MAC Address: 00:16:b6:f7:1d:51
- First-Hop Router MAC Address: 00:16:b6:f4:eb:a8
- Source IP Address: 192.168.1.109
- Destination IP Address: 128.119.245.12

Description: I examined Frame 474, which contains the TCP SYN segment initiating a connection to the web server. The IEEE 802.11 header identified the wireless host as the transmitter, the access point as the receiver, and the first-hop router as the destination. The IPv4 header identified the source and destination IP addresses.

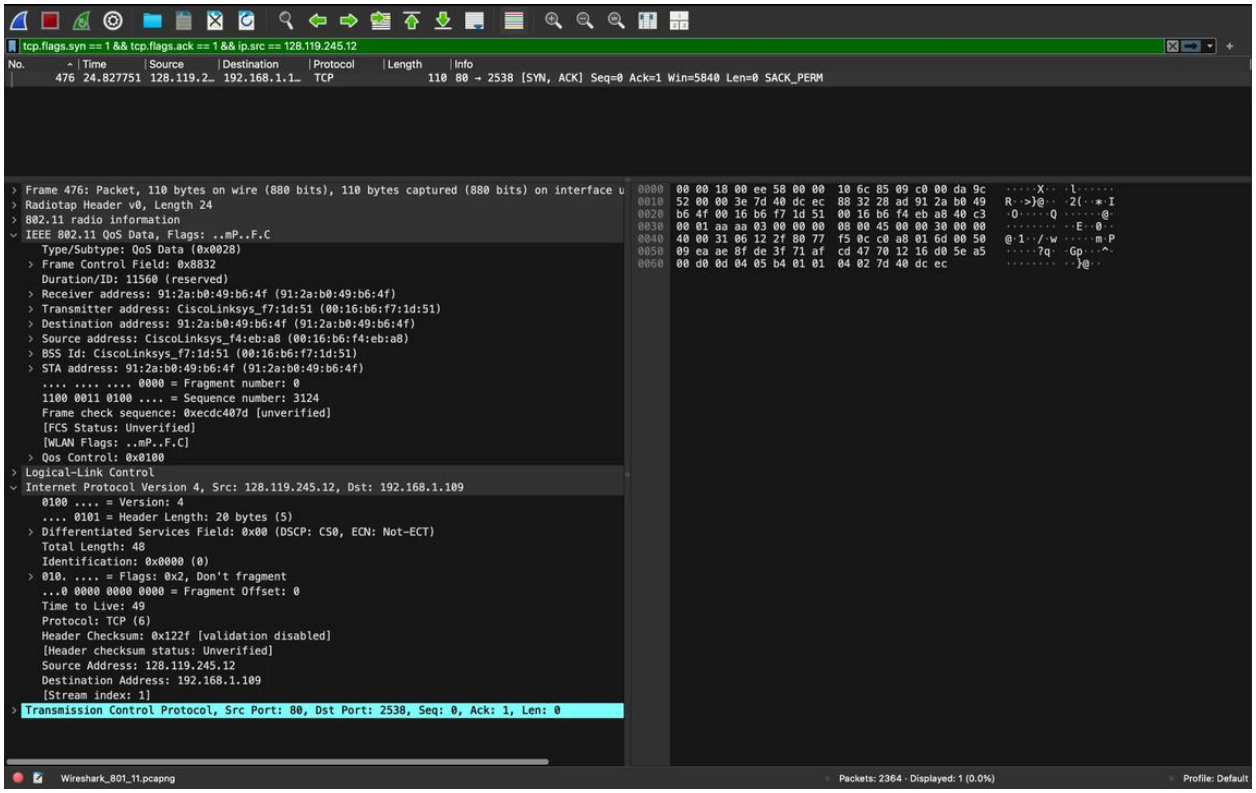
Question 9:



Answer: The destination IP address 128.119.245.12 corresponds to the destination web server which is gaia.cs.umass.edu.

Description: I looked at the IPv4 header of the TCP SYN packet in Frame 474, and although the packet travels through the wireless access point and first-hop router, the destination IP address identifies the web server that the wireless host is attempting to contact.

Question 10:

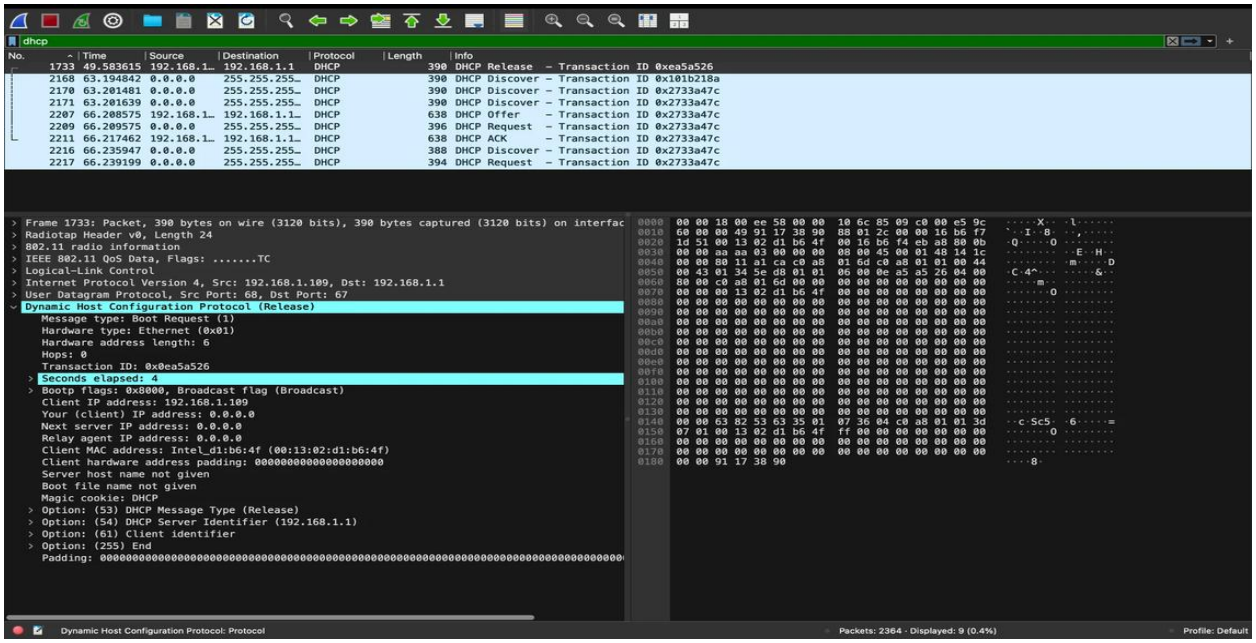


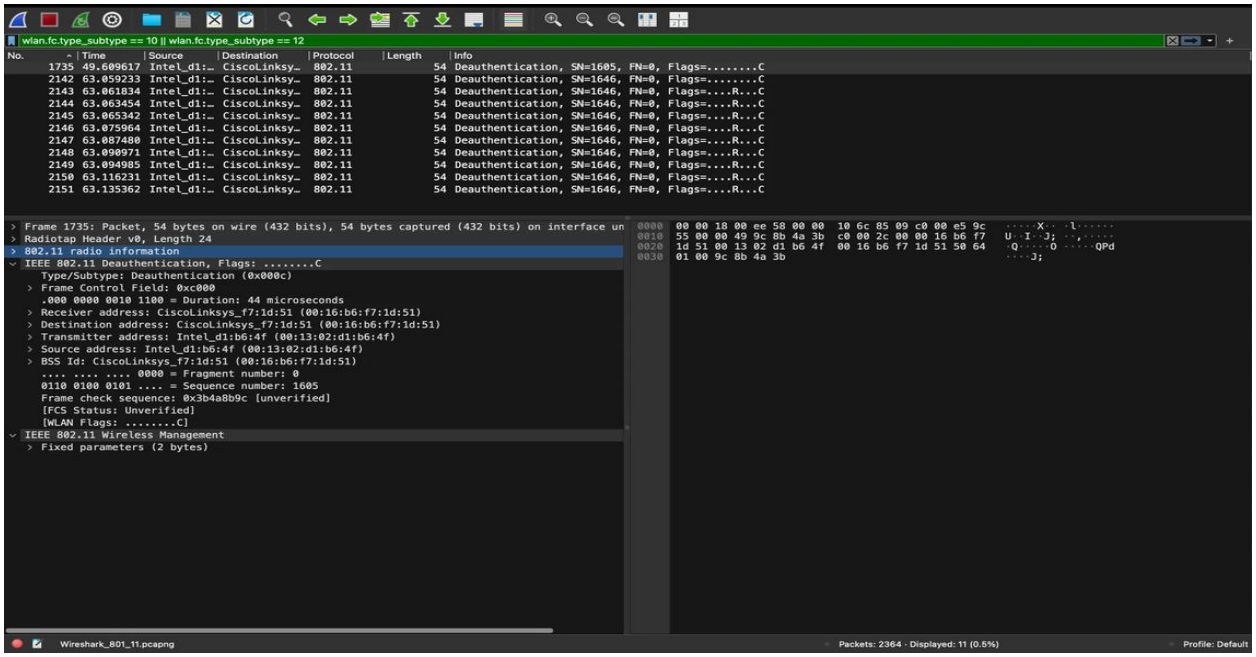
Answer: No, the transmitting MAC address in the wireless frame does not correspond to the IP address of the original TCP segment sender.

- Receiver/Destination MAC Address: 91:2a:b0:49:b6:4f
- Access Point MAC Address: 00:16:b6:f7:1d:51
- First-Hop Router MAC Address: 00:16:b6:f4:eb:a8
- Wireless Host MAC Address (previously identified): 00:13:02:d1:b6:4f

Description: I examined Frame 476 at timestamp 24.827751 seconds, which contains the TCP SYN-ACK response. The IPv4 header shows that the segment originated from the web server at 128.119.245.12, while the 802.11 header identifies the access point as the wireless frame's transmitter. This shows that MAC addresses identify local link-layer communication, whereas IP addresses identify network-layer endpoints. Additionally, the receiver MAC address recorded in Frame 476 also differs from the wireless host's MAC address identified in Frame 474.

Question 11:



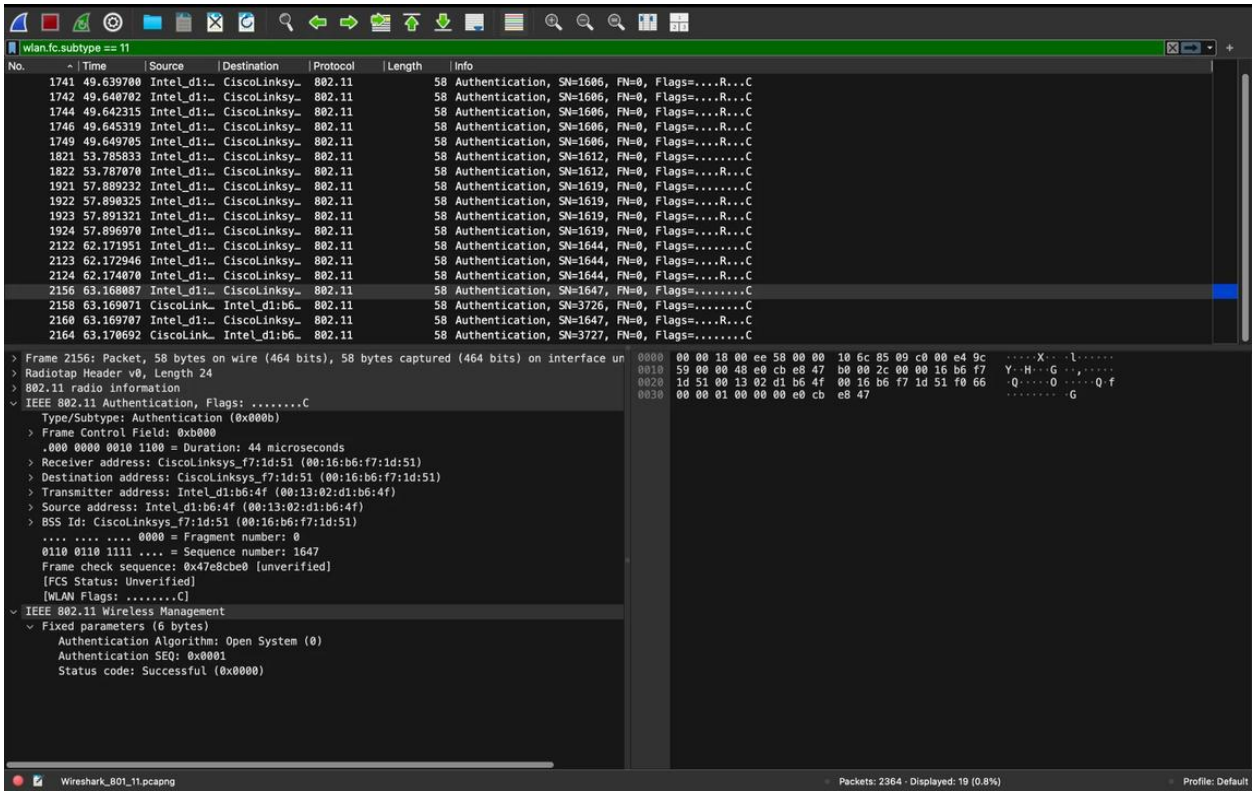


Answer: The wireless host performs two actions to disconnect from the 30 Munroe St access point:

- DHCP Release: At approximately 49.5836 seconds, the host releases its assigned IP address.
- 802.11 Deauthentication: At approximately 49.6096 seconds, the host sends a Deauthentication frame to the access point.

Description: I examined Frame 1733 and identified a DHCP Release message sent from the wireless host to the DHCP server. I then examined Frame 1735 and identified an IEEE 802.11 Deauthentication frame sent by the host to the access point. These actions release the IP address assignment and terminate the authenticated wireless connection.

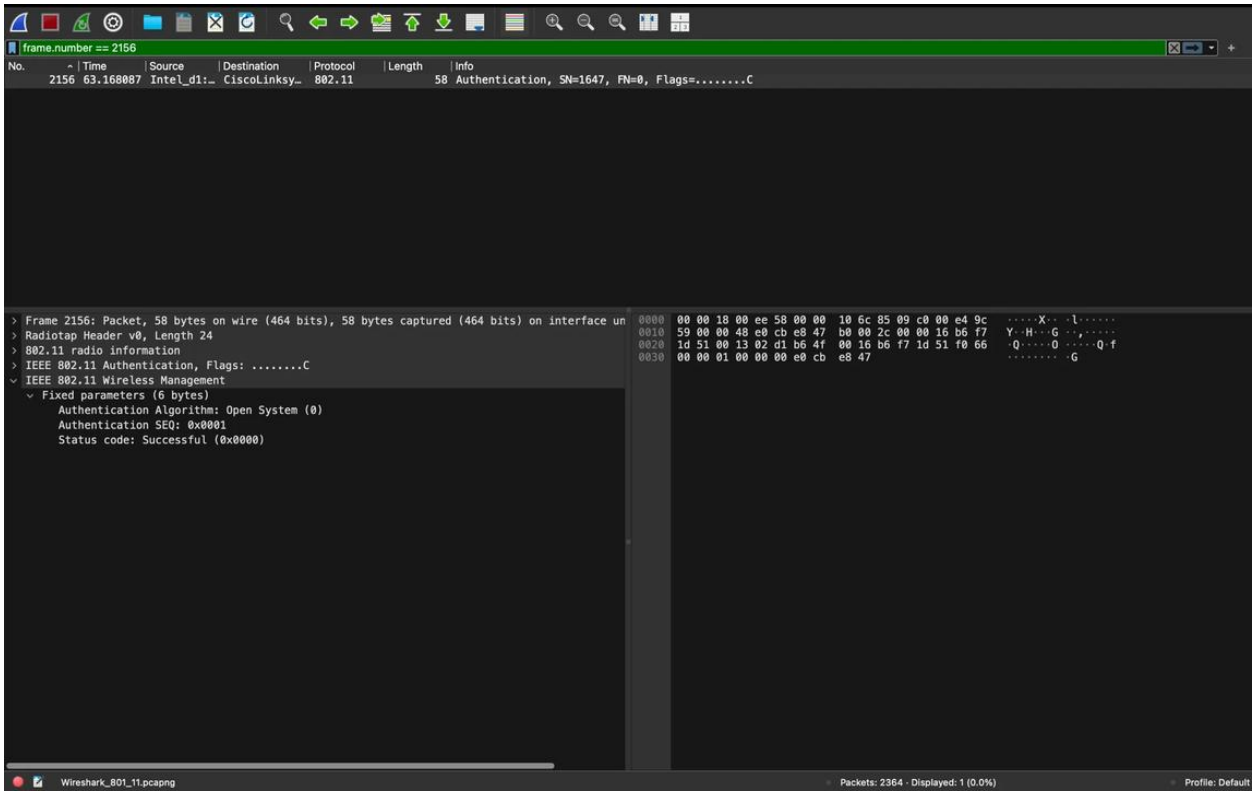
Question 12:



Answer: The wireless host requests Open System Authentication (0).

Description: I examined Frame 2156 at timestamp 63.168087 seconds. Under the fixed parameters in the IEEE 802.11 Wireless Management tab, the Authentication Algorithm field indicated Open System Authentication, identifying the authentication method requested by the host.

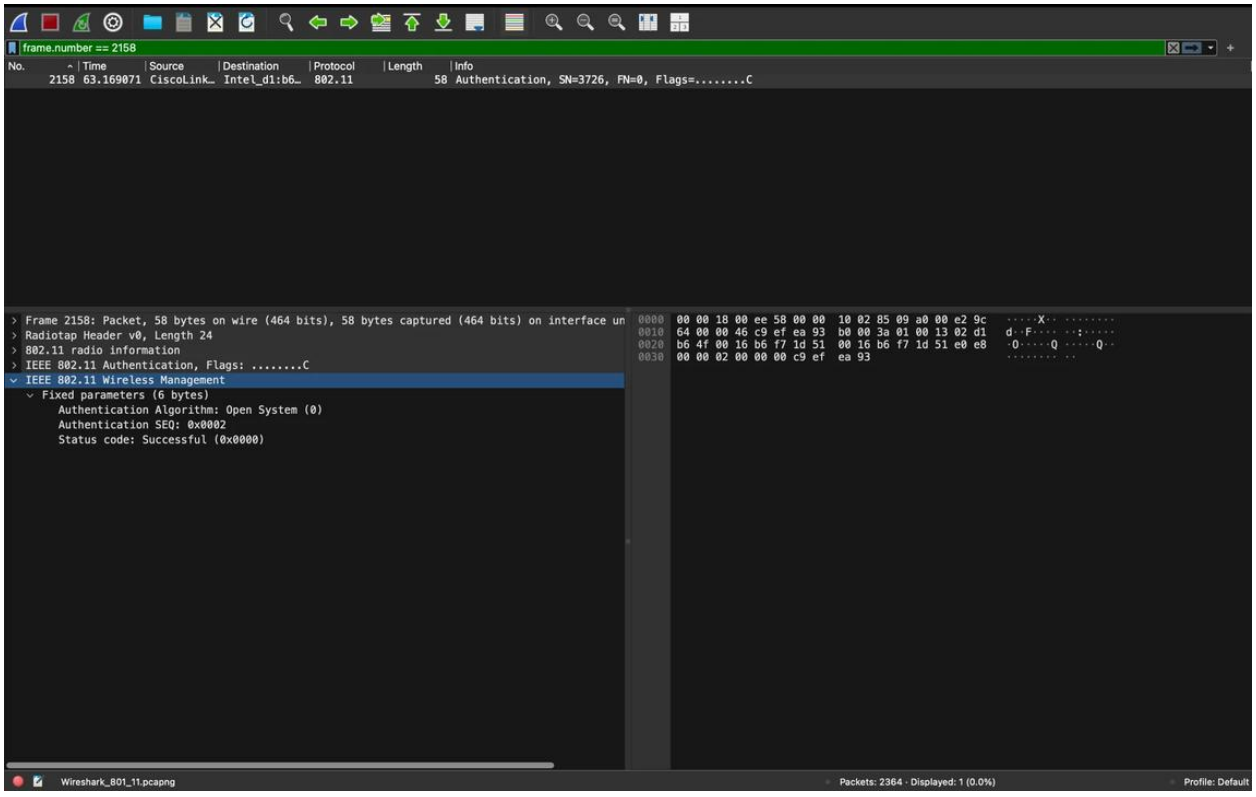
Question 13:



Answer: The Authentication SEQ value is 0x0001 (1).

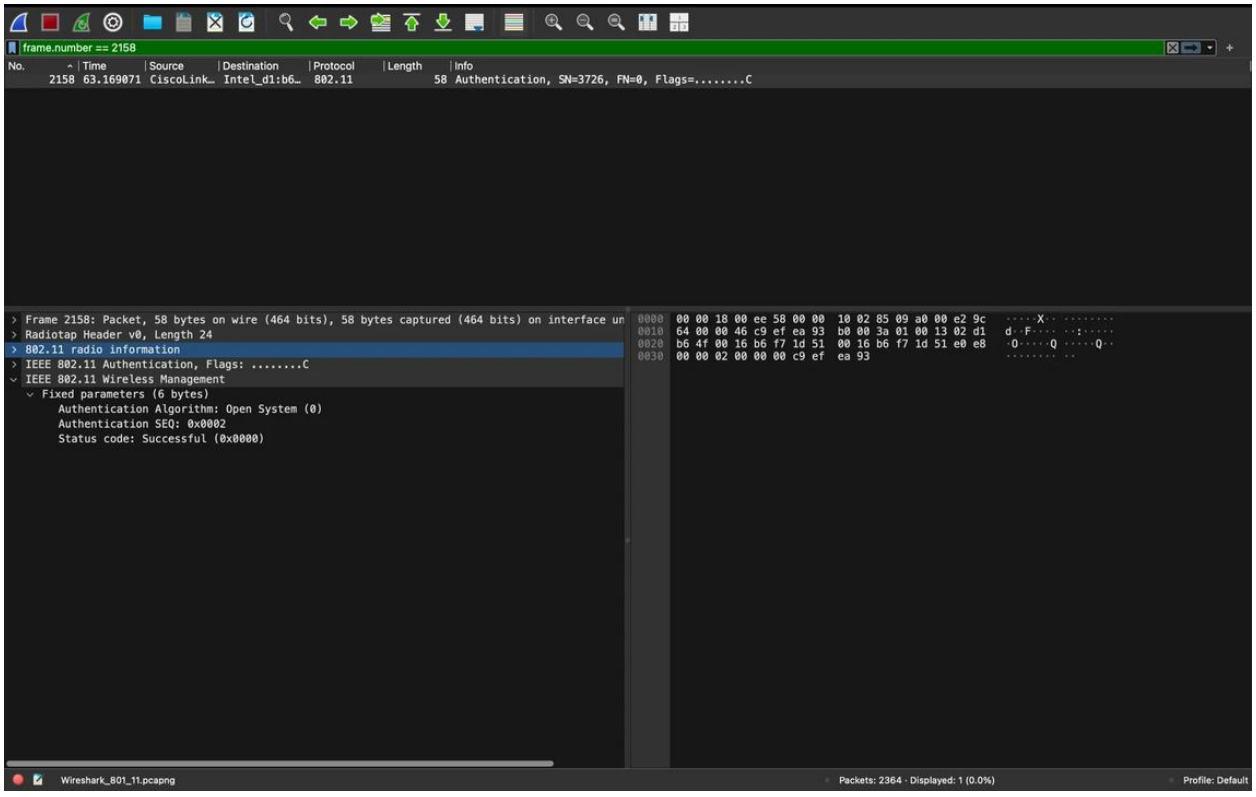
Description: I examined the Authentication Request in Frame 2156 and located the Authentication SEQ field under Fixed parameters. The value 0x0001 indicates the first message in the authentication exchange between the wireless host and access point.

Question 14:



Answer: Yes, the access point accepted the authentication method requested by the wireless host.
Description: I examined the Authentication Response in Frame 2158 at timestamp 63.169071 seconds. The Status Code field displayed Successful (0x0000), which confirms that the access point accepted the host's Open System Authentication request.

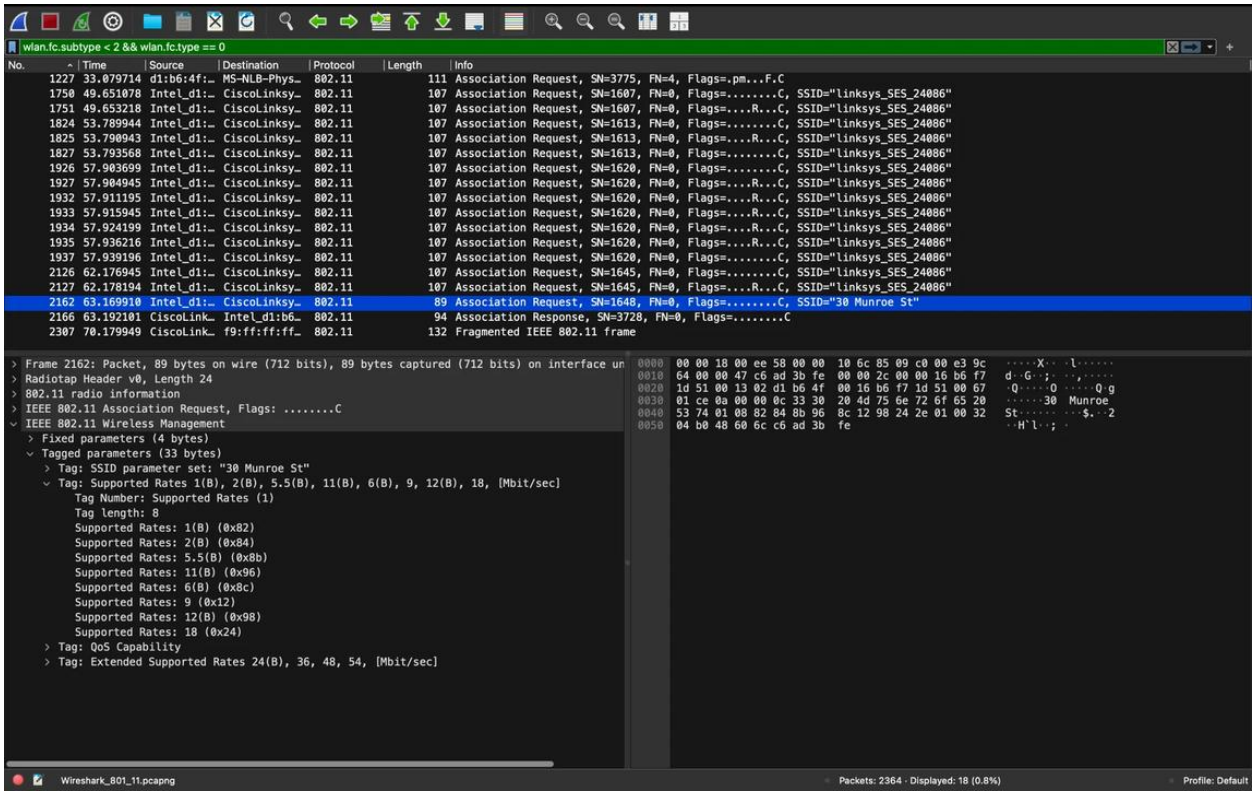
Question 15:



Answer: The Authentication SEQ value is 0x0002 (2).

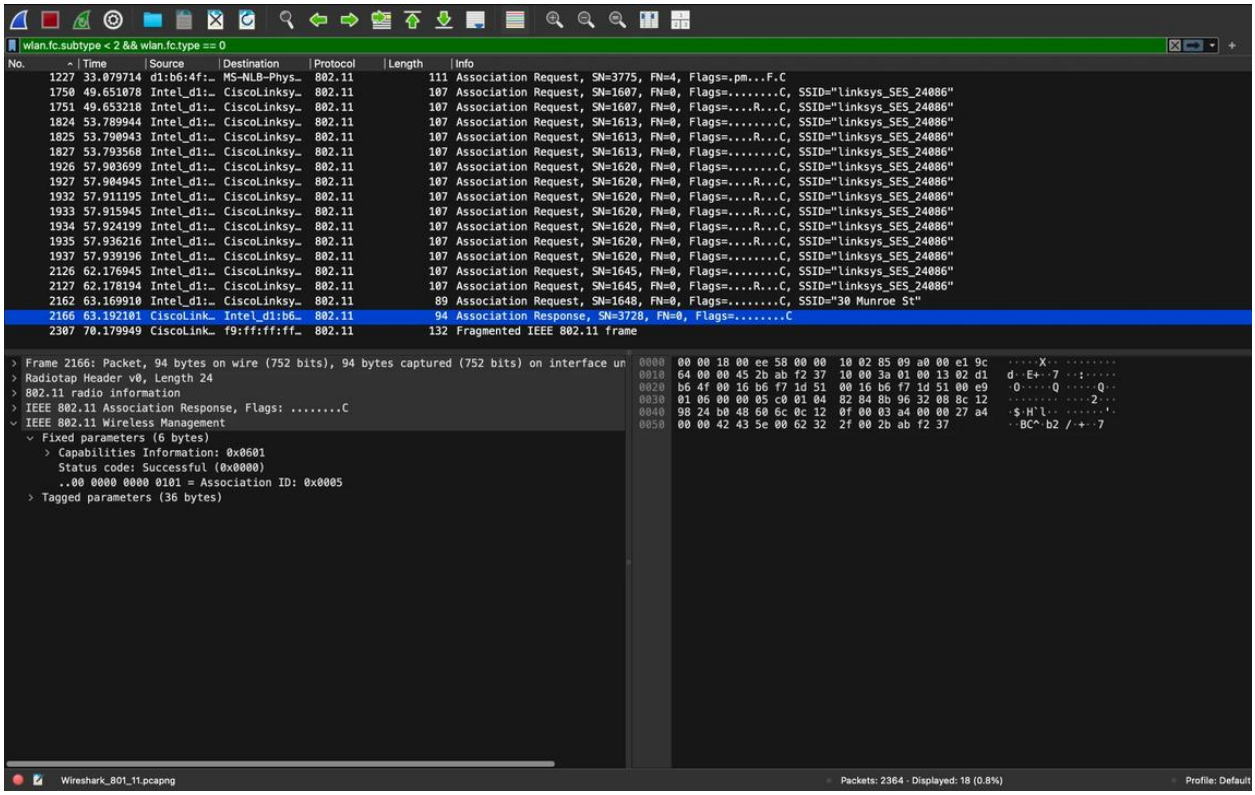
Description: I examined the Authentication Response in Frame 2158 and located the Authentication SEQ field under Fixed parameters. The value 0x0002 identifies the second message in the authentication exchange, which is transmitted by the access point to the wireless host.

Question 16:



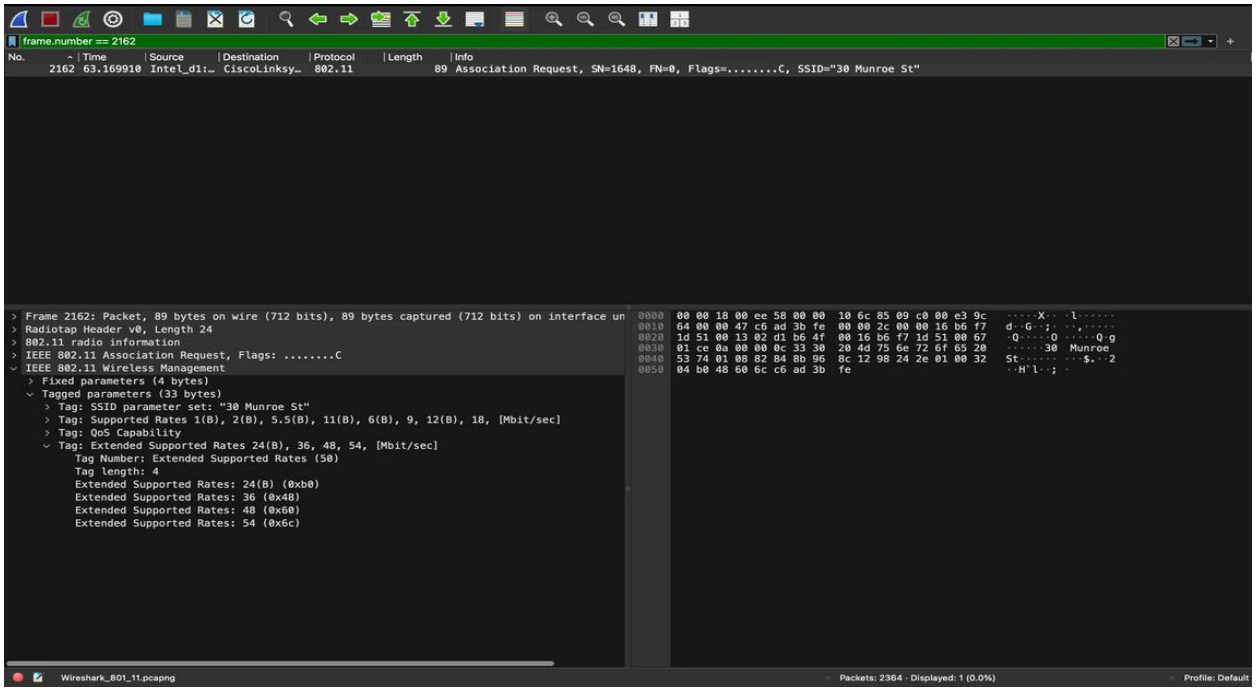
Answer: The supported rates advertised by the wireless host are 1, 2, 5.5, 11, 6, 9, 12, and 18 Mbps.
Description: I examined the Association Request in Frame 2162 at timestamp 63.169910 seconds. The Supported Rates field listed eight wireless data rates under the Tagged parameters in the IEEE 802.11 Wireless Management tab.

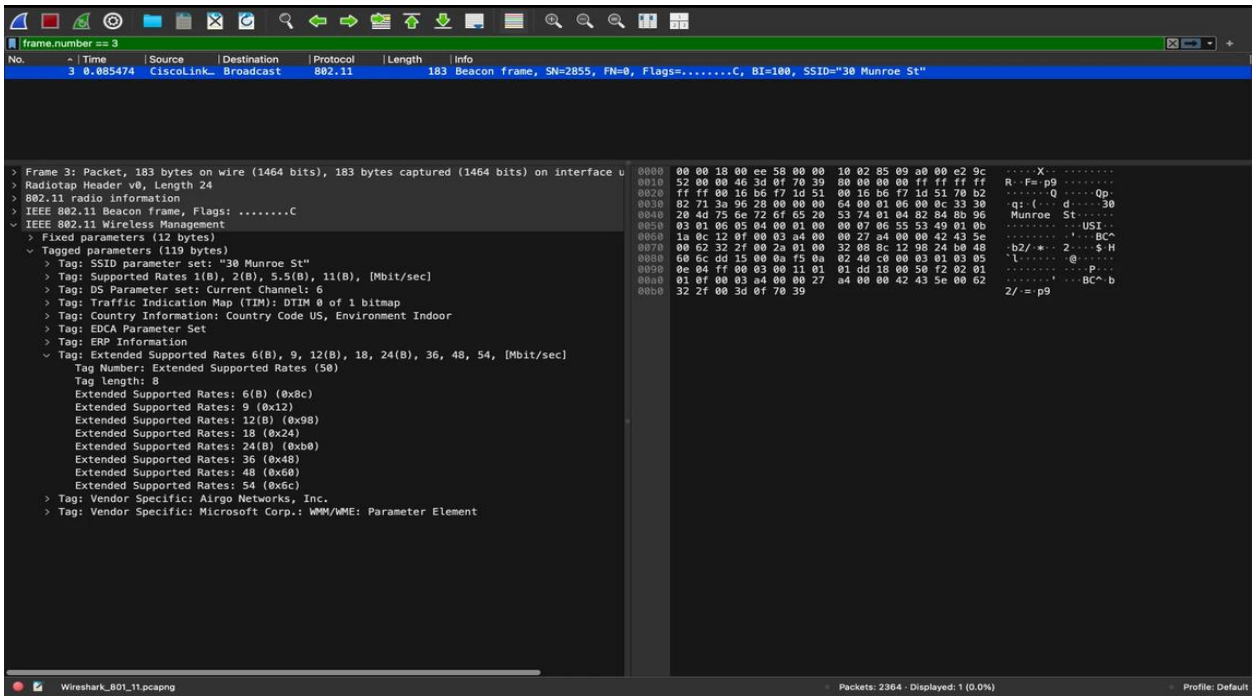
Question 17:



Answer: The Association Response indicates a successful association.
Description: I examined Frame 2166 at timestamp 63.192101 seconds. The Status Code displayed Successful (0x0000) in the Fixed parameters under the IEEE 802.11 Wireless Management tab, which confirms that the access point accepted the wireless host's Association Request.

Question 18:





Answer: Yes, the fastest Extended Supported Rate offered by the wireless host matches the fastest Extended Supported Rate supported by the access point. Both advertise a maximum rate of 54 Mbps.

Description: I examined the Extended Supported Rates field in the wireless host's Association Request (Frame 2162) and the 30 Munroe St access point's beacon frame (Frame 3). Both frames advertised 54 Mbps as their highest extended supported data rate, confirming that the host and access point support the same maximum advertised rate.